

# THE IVAN–WANG CONJECTURE ON INDUCED $\mathcal{N}$ -SATURATION FOR $1 \leq N \leq 5$

ABSTRACT. Let  $\mathcal{N}$  be the four-element poset with relations  $a < b$ ,  $c < b$ , and  $c < d$ . We prove

$$\text{sat}^*(n, \mathcal{N}) = 2n \quad (1 \leq n \leq 5).$$

The cases  $n = 1, 2$  are immediate. For  $n = 3, 4, 5$ , the proof is a complete exact enumeration after reducing induced saturation to maximal independence in a 4-uniform hypergraph. The numbers of labeled minimum families are 9, 60, and 450; up to ground-set permutations and complementation, they form 2, 4, and 7 classes. Thus Conjecture 16 of Ivan and Wang is verified in the first three nonvacuous dimensions. No claim is made for  $n \geq 6$ .

## 1. RESULT

Let  $\mathcal{B}_n = 2^{[n]}$ , ordered by inclusion. A family  $\mathcal{F} \subseteq \mathcal{B}_n$  is *induced- $\mathcal{N}$ -saturated* if it contains no induced copy of  $\mathcal{N}$ , while  $\mathcal{F} \cup \{S\}$  contains one for every  $S \in \mathcal{B}_n \setminus \mathcal{F}$ . The minimum size of such a family is  $\text{sat}^*(n, \mathcal{N})$ .

Ferrara et al. established the upper bound

$$\text{sat}^*(n, \mathcal{N}) \leq 2n \quad (n \geq 3)$$

by an explicit construction [1, Proposition 2]. Ivan and Wang proved  $\text{sat}^*(n, \mathcal{N}) \geq (n+6)/4$  [2, Theorem 12]. They conjectured that the upper bound is always exact [2, Conjecture 16].

The nearest neighbouring case is the four-element poset  $2C_2$ , two disjoint 2-chains: there the upper bound  $2n$  is likewise known, and Martin and Veldt raised the lower bound to  $3n/2 + 1/2$  and exhibited many induced- $2C_2$ -saturated families of size  $2n$  [3]. For  $\mathcal{N}$  the present note determines the value exactly in the smallest dimensions instead of bounding it.

**Theorem 1.** *For every  $1 \leq n \leq 5$ ,*

$$\text{sat}^*(n, \mathcal{N}) = 2n.$$

*For  $n = 3, 4, 5$ , the numbers of labeled minimum families are, respectively,*

$$9, \quad 60, \quad 450.$$

*Under ground-set permutations and complementation, these minimum families form, respectively, 2, 4, and 7 classes.*

The theorem is a finite verification of the Ivan–Wang conjecture, not a proof of its general form.

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## 2. EXACT REDUCTION

For a collection  $Q$  of subsets, let  $G(Q)$  be its comparability graph: two vertices are adjacent exactly when the corresponding sets are comparable by inclusion.

**Lemma 2.** *Every induced- $\mathcal{N}$ -saturated family in  $\mathcal{B}_n$  contains  $\emptyset$  and  $[n]$ . Moreover, four distinct subsets induce  $\mathcal{N}$  if and only if their comparability graph is a path  $P_4$ .*

*Proof.* Every element of  $\mathcal{N}$  is incomparable with another element. Since  $\emptyset$  and  $[n]$  are comparable with every member of  $\mathcal{B}_n$ , neither can belong to an induced copy of  $\mathcal{N}$ . If either were absent from a saturated family, adding it would create no induced copy, a contradiction.

The comparability graph of  $\mathcal{N}$  is  $P_4$ . Conversely, orient every edge of a comparability graph  $P_4$  from the smaller set to the larger set. Two consecutive edges cannot point in the same direction, since transitivity would create a chord. Hence the orientation alternates, giving  $\mathcal{N}$  or its dual. These two posets are isomorphic.  $\square$

Set

$$V_n = \mathcal{B}_n \setminus \{\emptyset, [n]\},$$

and let  $\mathcal{H}_n = (V_n, \mathcal{E}_n)$  be the 4-uniform hypergraph whose edges are the four-subsets inducing  $\mathcal{N}$ .

**Lemma 3.** *For  $X \subseteq V_n$ , the family  $\{\emptyset, [n]\} \cup X$  is induced- $\mathcal{N}$ -saturated if and only if  $X$  is a maximal independent set of  $\mathcal{H}_n$ .*

*Proof.* By Lemma 2, the two extreme sets cannot occur in an induced copy of  $\mathcal{N}$ . Thus  $\{\emptyset, [n]\} \cup X$  is induced- $\mathcal{N}$ -free exactly when  $X$  contains no edge of  $\mathcal{H}_n$ . For  $v \in V_n \setminus X$ , adding  $v$  creates an induced copy exactly when some edge of  $\mathcal{H}_n$  is contained in  $X \cup \{v\}$ . This is precisely maximal independence.  $\square$

The number of hyperedges also has a closed form, providing an independent check on their generation.

**Lemma 4.** *For every  $n \geq 1$ ,*

$$|\mathcal{E}_n| = 8^n - 3 \cdot 7^n + 3 \cdot 6^n - 5^n.$$

*Proof.* First, an induced copy of  $\mathcal{N}$  determines its roles uniquely, because  $\text{Aut}(\mathcal{N})$  is trivial: an automorphism preserves the set of minimal elements  $\{a, c\}$  and the number of upper covers, and  $a$  is the unique minimal element with one upper cover while  $c$  is the unique minimal element with two, so  $a$  and  $c$  are fixed, and then  $b$  (the upper cover of  $a$ ) and  $d$  (the remaining element) are fixed as well. Hence a four-element subset of  $\mathcal{B}_n$  inducing  $\mathcal{N}$  admits exactly one labelling  $A, B, C, D$  with  $A < B$ ,  $C < B$ , and  $C < D$ , and counting induced copies of  $\mathcal{N}$  is the same as counting ordered quadruples  $(A, B, C, D)$  of subsets of  $[n]$  that induce  $\mathcal{N}$  in these roles. By Lemma 2 no

induced copy contains  $\emptyset$  or  $[n]$ , so all four sets lie in  $V_n$  and this common count is  $|\mathcal{E}_n|$ .

So let  $(A, B, C, D)$  satisfy  $A \subseteq B$ ,  $C \subseteq B$ , and  $C \subseteq D$ . For each coordinate, the membership pattern  $(\mathbf{1}_A, \mathbf{1}_B, \mathbf{1}_C, \mathbf{1}_D)$  is then one of

$$0000, 0001, 0100, 0101, 1100, 1101, 0111, 1111,$$

these being exactly the eight of the sixteen patterns compatible with the three inclusions. The quadruple induces  $\mathcal{N}$  in the roles  $A, B, C, D$  if and only if each of 0001, 1100, and 0111 occurs. Indeed, these patterns respectively witness  $D \not\subseteq B$ ,  $A \not\subseteq D$ , and  $C \not\subseteq A$ ; together with the three prescribed inclusions, they also force all strictness and all remaining incomparabilities. Inclusion–exclusion over the three required patterns gives the formula.  $\square$

### 3. ENUMERATION AND SYMMETRY CLASSES

For a triple  $T \in \binom{V_n}{3}$ , define its completion mask

$$M_n(T) = \{v \in V_n : T \cup \{v\} \in \mathcal{E}_n\}.$$

For each candidate  $X \subseteq V_n$ , the verifier forms

$$C_n(X) = \bigcup_{T \in \binom{X}{3}} M_n(T).$$

Then  $X$  is independent exactly when  $C_n(X) \cap X = \emptyset$ , and it is maximal independent exactly when, in addition,  $C_n(X) \cup X = V_n$ .

For each  $n \in \{3, 4, 5\}$ , every candidate with  $|X| \leq 2n - 2$  was tested. The number tested was

$$\sum_{k=0}^{2n-2} \binom{2n-2}{k}.$$

All operations were exact integer and bit-mask operations.

The enumeration records the size of every maximal independent set it finds. Writing  $m_n(k)$  for the number of maximal independent sets of  $\mathcal{H}_n$  with  $|X| = k$ , the tested range  $0 \leq k \leq 2n - 2$  gives

$$(1) \quad \begin{aligned} (m_3(k))_{k=0}^4 &= (0, 0, 0, 0, 9), \\ (m_4(k))_{k=0}^6 &= (0, 0, 0, 0, 0, 0, 60), \\ (m_5(k))_{k=0}^8 &= (0, 0, 0, 0, 0, 0, 0, 0, 450). \end{aligned}$$

In particular  $m_n(k) = 0$  for every  $k < 2n - 2$ : no maximal independent set of  $\mathcal{H}_n$  is smaller than  $2n - 2$ . The final entries 9, 60, 450 are the totals in the last column of Table 1.

*Proof of Theorem 1.* For  $n = 1, 2$ , the lattice  $\mathcal{B}_n$  contains no induced copy of  $\mathcal{N}$ , so its only induced- $\mathcal{N}$ -saturated family is  $\mathcal{B}_n$  itself. Hence  $\text{sat}^*(1, \mathcal{N}) = 2$  and  $\text{sat}^*(2, \mathcal{N}) = 4$ .

Now let  $n \in \{3, 4, 5\}$ . By Lemma 3, an induced- $\mathcal{N}$ -saturated family of size at most  $2n$  corresponds exactly to a maximal independent set  $X \subseteq V_n$  with

| $n$ | $ V_n $ | $ \mathcal{E}_n $ | tested    | independent | maximal |
|-----|---------|-------------------|-----------|-------------|---------|
| 3   | 6       | 6                 | 57        | 51          | 9       |
| 4   | 14      | 156               | 6 476     | 3 110       | 60      |
| 5   | 30      | 2 550             | 8 656 937 | 1 690 127   | 450     |

TABLE 1. Complete enumeration for  $|X| \leq 2n - 2$ . Every maximal independent set counted in the last column has size exactly  $2n - 2$ ; the sizes are broken out in (1).

$|X| \leq 2n - 2$ . By (1) there is none with  $|X| < 2n - 2$ , so  $\text{sat}^*(n, \mathcal{N}) \geq 2n$ , and the number with  $|X| = 2n - 2$  is 9, 60, and 450 respectively.

The upper bound is exhibited on this range by the enumeration itself: each representative  $X$  of Table 2 has  $|X| = 2n - 2$ , so  $\{\emptyset, [n]\} \cup X$  is an induced- $\mathcal{N}$ -saturated family of size  $2n$  by Lemma 3. It also follows for all  $n \geq 3$  from the construction of Ferrara et al. [1, Proposition 2].  $\square$

It remains to record the symmetry classification. Let

$$\Gamma_n = S_n \times C_2,$$

where  $S_n$  permutes the ground set and the nontrivial element of  $C_2$  sends each subset to its complement. Complementation reverses inclusion but preserves induced- $\mathcal{N}$ -saturation because  $\mathcal{N}$  is self-dual.

To define canonical representatives, encode a subset  $S \subseteq [n]$  by

$$\mu(S) = \sum_{i \in S} 2^{i-1},$$

sort each family by increasing  $\mu$ , and take the lexicographically least image under  $\Gamma_n$ . In Table 2, a word such as 125 denotes  $\{1, 2, 5\}$ ; only the members of  $X$  are displayed.

**Proposition 5.** *For  $n = 3, 4, 5$ , the minimum families have the orbit representatives and orbit sizes listed in Table 2. In particular, there are 2, 4, and 7 orbits, respectively.*

*Proof.* For each minimum family, the classifier applies all  $2 \cdot n!$  elements of  $\Gamma_n$  and selects the canonical image defined above. Grouping equal images gives Table 2. The orbit sizes sum to 9, 60, and 450, agreeing with the complete labeled census.  $\square$

This is a classification under the natural symmetries and duality of the Boolean lattice, not under arbitrary abstract poset isomorphism. It gives a finite partial answer to Ivan and Wang’s structural Question 15 [2].

#### COMPUTATIONAL SPECIFICATION

Nothing outside this note is needed. The computation is specified completely by the following five steps, each a finite exact computation that a

| $n$ | orbit size | canonical representative $X$       |
|-----|------------|------------------------------------|
| 3   | 6          | 1, 2, 12, 3                        |
| 3   | 3          | 1, 2, 13, 23                       |
| 4   | 24         | 1, 2, 12, 3, 123, 4                |
| 4   | 6          | 1, 2, 12, 3, 4, 34                 |
| 4   | 24         | 1, 2, 12, 3, 124, 34               |
| 4   | 6          | 1, 2, 12, 123, 124, 34             |
| 5   | 120        | 1, 2, 12, 3, 123, 4, 1234, 5       |
| 5   | 60         | 1, 2, 12, 3, 123, 4, 5, 45         |
| 5   | 120        | 1, 2, 12, 3, 123, 4, 1235, 45      |
| 5   | 60         | 1, 2, 12, 3, 123, 1234, 1235, 45   |
| 5   | 30         | 1, 2, 12, 3, 4, 34, 1234, 5        |
| 5   | 30         | 1, 2, 12, 3, 4, 34, 125, 345       |
| 5   | 30         | 1, 2, 12, 123, 1234, 1235, 45, 345 |

TABLE 2. Minimum families up to ground-set permutations and complementation.

reader can reimplement from this description alone; Tables 1 and 2 and (1) are its entire output.

- (1) *Ground set.* Identify  $S \subseteq [n]$  with the bit-mask  $\mu(S)$ , so that  $V_n$  is the set of integers  $1, \dots, 2^n - 2$  and  $S \subseteq T$  holds exactly when the bitwise conjunction of  $\mu(S)$  and  $\mu(T)$  is  $\mu(S)$ .
- (2) *Edges.* For every 4-subset of  $V_n$ , decide membership in  $\mathcal{E}_n$  by testing whether its comparability graph is a path on four vertices (Lemma 2); equivalently, by testing the 24 bijections onto  $\mathcal{N}$  for an isomorphism of induced subposets. The resulting count is compared with Lemma 4.
- (3) *Masks.* For each  $T \in \binom{V_n}{3}$ , store  $M_n(T)$  as a bit-mask over  $V_n$ ; since  $|V_n| \leq 30$ , one machine word suffices.
- (4) *Census.* Enumerate every  $X \subseteq V_n$  with  $|X| \leq 2n - 2$ , accumulate  $C_n(X)$  as the disjunction of the masks of the triples of  $X$ , and record  $X$  as independent when  $C_n(X) \cap X = \emptyset$  and as maximal independent when in addition  $C_n(X) \cup X = V_n$ . This produces the counts of Table 1 and the size distribution (1).
- (5) *Classes.* Realise  $\Gamma_n$  as  $2 \cdot n!$  permutations of  $V_n$ : a ground-set permutation permutes the bits of  $\mu$ , and complementation is  $\mu \mapsto 2^n - 1 - \mu$ . Apply all of them to each recorded minimum family, sort each image by increasing  $\mu$ , and keep the lexicographically least word. Families with equal canonical images form the classes of Table 2.

Everything is exact integer and bit-mask arithmetic. There is no randomization, floating-point arithmetic, external solver, or tolerance parameter, and no step depends on the order of enumeration.

## EXACT VERIFICATION

Every value printed above was recomputed independently: a second program, written from the definitions of this note rather than from the first implementation, was run on a separate machine. It regenerates the induced copies of  $\mathcal{N}$  in  $\mathcal{B}_n$  directly from the definition, confirms the closed form and the role uniqueness of Lemma 4, reproduces the census of Table 1 entire by two independent methods (an unpruned scan and a pruned depth-first search), with all 57, 6 476 and 8 656 937 candidates tested and nothing sampled or truncated, checks each of the 9, 60 and 450 minimum families against the definition of induced- $\mathcal{N}$ -saturation, and recovers the orbits and orbit sizes of Table 2 by closing the census under  $\Gamma_n$  instead of by canonical forms. All 105 of its checks passed and every recomputed value agreed with the value printed here. For  $n \leq 4$  it also obtains  $\text{sat}^*(n, \mathcal{N})$  with no lemma at all, by scanning all  $2^{2^n}$  subfamilies of  $\mathcal{B}_n$ ; that scan is out of reach at  $n = 5$ , where the passage from the census to  $\text{sat}^*(5, \mathcal{N}) = 10$  therefore rests on Lemma 3, whose computational content, that no induced copy of  $\mathcal{N}$  meets  $\emptyset$  or  $[n]$ , is checked at  $n = 5$  and whose full set equality is checked at  $n = 3, 4$ , and nothing is checked for  $n \geq 6$  beyond the edge count of Lemma 4. That second program and the transcript of its run accompany this note; the first implementation is not included here. The specification above is what a reader should reimplement. An independent recomputation that agrees is evidence; the proof is the enumeration specified above.

## STATUS AND SCOPE

Theorem 1 settles Conjecture 16 of Ivan and Wang for  $n \leq 5$  and leaves it open for every  $n \geq 6$ ; it gives no information about the asymptotic behaviour of  $\text{sat}^*(n, \mathcal{N})$ , about the lower bound of [2, Theorem 12], or about posets other than  $\mathcal{N}$ . Proposition 5 classifies under  $\Gamma_n$  only. We have not found a previous determination of  $\text{sat}^*(n, \mathcal{N})$  for  $3 \leq n \leq 5$ : a search of the poset-saturation literature, including [1, 2, 3], turned up the bounds recalled at the start of this note but no exact value in these dimensions. That search was not exhaustive and has not been through external review, so any earlier determination we have missed takes precedence over this note.

## REFERENCES

- [1] M. Ferrara, B. Kay, L. Kramer, R. R. Martin, B. Reiniger, H. C. Smith, and E. Sullivan, *The saturation number of induced subposets of the Boolean lattice*, Discrete Math. **340** (2017), no. 10, 2479–2487, doi:10.1016/j.disc.2017.06.010.
- [2] M.-R. Ivan and N. Wang, *Linear saturation for  $\mathcal{N}$  via butterflies*, arXiv:2511.08965v2 [math.CO], 2026.
- [3] R. R. Martin and N. Veldt, *Induced saturation of the poset  $2C_2$* , Ann. Comb. **30** (2026), no. 1, 77–90, doi:10.1007/s00026-025-00764-z; arXiv:2408.14648.